

SOC ANALYST PORTFOLIO CASE STUDY

Investigating a Failed Privilege-Escalation Attempt

Wazuh detection and triage of three failed sudo authentication attempts

RYAN BRYANT | SOC / CYBERSECURITY ANALYST CANDIDATE | AUGUST 2026

OUTCOME Generated a controlled failed sudo request, verified the local authentication record, and investigated Wazuh rule 5404 at Level 10 with MITRE ATT&CK T1548.003 context. No privileged command executed and no system change occurred.

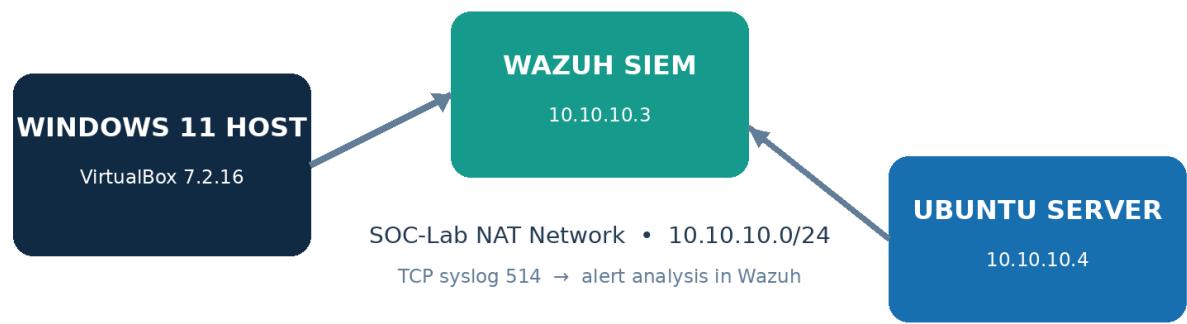


Figure 1. Home-lab architecture and authentication telemetry flow.

Portfolio note: Authorized local simulation only. The requested command was intentionally harmless and did not execute. AI assistance is disclosed within this report.

1. Executive Summary

I tested Wazuh's ability to detect repeated failed sudo authentication on the Ubuntu endpoint. After documenting the socstudent identity and sudo-group membership, validating rsyslog, and confirming the TCP/514 connection to Wazuh, I invalidated cached sudo authentication and requested the harmless `/usr/bin/true` command as root.

I intentionally entered an incorrect password three times. sudo denied the request and returned control to the unprivileged shell. The authentication log recorded the source account, destination account, terminal, working directory, command, and failure count. Wazuh decoded the event as sudo activity and generated rule 5404 at Level 10: Three failed attempts to run sudo. The alert mapped to MITRE ATT&CK T1548.003, Sudo and Sudo Caching.

Project dimension	Implementation
Role performed	Detection tester and Tier 1 incident analyst
Telemetry source	Ubuntu sudo authentication log
Source / target identity	socstudent / root
Requested command	<code>/usr/bin/true</code>
Preventive result	Authentication denied after three failures
Detection result	Wazuh rule 5404; Level 10
MITRE context	T1548.003 — Sudo and Sudo Caching
Disposition	True positive detection; expected authorized lab activity

2. Objectives

- Document the authorized user and privilege baseline.
- Generate a controlled sudo authentication failure without executing a privileged action.
- Verify the local log before relying on the SIEM alert.
- Investigate decoded identities, command context, rule severity, and MITRE mapping.
- Confirm no root session, persistence, or system modification occurred.

3. Scope and Safety Controls

Control	Implementation
Authorized endpoint	linux-webserver01 at 10.10.10.4
Authorized account	socstudent; member of sudo group
Command safety	<code>/usr/bin/true</code> performs no modification
Failure limit	Exactly three intentionally incorrect password entries
Telemetry	rsyslog TCP/514 to Wazuh 10.10.10.3
Stop condition	No correct credential entered; no privileged shell requested

4. Implementation Workflow

01 IDENTITY BASELINE

Recorded whoami and id output for socstudent and sudo-group membership.

02 PIPELINE VALIDATION

Confirmed rsyslog active and TCP/514 established to Wazuh.

03 CREDENTIAL RESET

Used sudo -k to invalidate any cached authentication.

04 CONTROLLED REQUEST

Requested /usr/bin/true with a unique sudo prompt.

05 FAILURE GENERATION

Entered an incorrect password three times.

06 LOCAL VERIFICATION

Confirmed auth.log recorded the failure count, USER=root, and COMMAND=/usr/bin/true.

07 SIEM SEARCH

Searched incorrect-password events and separated the failed attempt from the later verification command.

08 EVENT INVESTIGATION

Captured rule 5404, Level 10, decoded identities, command fields, and MITRE T1548.003.

09 DISPOSITION

Confirmed the attempt failed and no privileged change occurred.

5. Controlled Test Command

```
sudo -k
```

```
sudo -p 'SOC-Lab-Sudo-Failure: ' /usr/bin/true
```

SAFETY BOUNDARY The command was selected because /usr/bin/true performs no system modification. Authentication failed before sudo could execute it.

6. Evidence: Identity and Telemetry Baseline

```
socstudent@linux-webserver01:~$ whoami
socstudent
socstudent@linux-webserver01:~$ id
uid=1000(socstudent) gid=1000(socstudent) groups=1000(socstudent),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),101(lxd)
socstudent@linux-webserver01:~$ systemctl is-active rsyslog
Command 'systemctl' not found, did you mean:
  command 'systemctl' from deb systemd (255.4-1ubuntu8.16)
  command 'systemctl' from deb systemctl (1.4.4181-1.1)
Try: sudo apt install <deb name>
socstudent@linux-webserver01:~$ ss -tn | grep '10.10.10.3:514'
ESTAB 0      0      10.10.10.4:32874    10.10.10.3:514
socstudent@linux-webserver01:~$ systemctl is-active rsyslog
active
socstudent@linux-webserver01:~$
```

Figure 2. User baseline and healthy syslog transport.

Baseline field	Observed value
Current user	socstudent
UID / GID	1000 / 1000
Relevant group	sudo (GID 27)
Rsyslog	active
Transport	10.10.10.4 to 10.10.10.3:514 ESTAB

7. Evidence: Failed Authentication and Local Record

```
socstudent@linux-webserver01:~$ sudo -k
socstudent@linux-webserver01:~$ sudo -p 'SOC-Lab-Sudo-Failure: ' /usr/bin/true
SOC-Lab-Sudo-Failure:
inSorry, try again.
SOC-Lab-Sudo-Failure:
Sorry, try again.
SOC-Lab-Sudo-Failure:
sudo: 3 incorrect password attempts
socstudent@linux-webserver01:~$
```

Figure 3. Controlled failed sudo authentication.

```
socstudent@linux-webserver01:~$ sudo grep -E 'authentication failure|incorrect password attempts' /var/log/auth.log | tail -n 10
2026-08-20T18:09:53.388623+00:00 linux-webserver01 sudo: socstudent : 3 incorrect password attempts ; TTY=tty1 ; PWD=/home/socstudent ; USER=root ; COMMAND=/usr/
bin/true
2026-08-20T18:17:40.567273+00:00 linux-webserver01 sudo: socstudent : TTY=tty1 ; PWD=/home/socstudent ; USER=root ; COMMAND=/usr/bin/grep -E 'authentication failu
re|incorrect password attempts' /var/log/auth.log
socstudent@linux-webserver01:~$
```

Figure 4. Local sudo authentication evidence.

Field	Observed value
Source account	socstudent
Destination account	root
TTY / directory	tty1 / /home/socstudent
Requested command	/usr/bin/true
Failure count	3
Execution result	Denied; command not executed

8. Evidence: Wazuh Event Investigation

W.

Discover

NewSaveOpenShareReportingInspect

wazuh-alerts-*

Search field names

Filter by type0

Selected fields

_source

Available fields

Aug 20, 2026 @ 14:09

predecoder.hostname: linux-webserver01

predecoder.program_name: sudo

predecoder.timestamp: Aug 20 18:09:53

input.type: log

agent.name: wazuh-server

agent.id: 000

manager.name: wazuh-server

data.srcuser: socstudent

data.dstuser: root

data.tty: tty1

data.pwd: /home/socstudent

data.command: /usr/bin/true

rule.mail: false

rule.level: 10

rule.pci_dss: 10.2.4, 10.2.5

rule.hipaa: 164.312.b

rule.tsc: CC6.1, CC

Expanded document

View surrounding documents

View single document

TableJSON

f _index	wazuh-alerts-4.x-2026.08.20
f agent.id	000
f agent.name	wazuh-server
f data.command	/usr/bin/true
f data.dstuser	root
f data.pwd	/home/socstudent
f data.srcuser	socstudent
f data.tty	tty1
f decoder.ftscomment	First time user executed the sudo command
f decoder.name	sudo
f decoder.parent	sudo
f full_log	Aug 20 18:09:53 linux-webserver01 sudo: socstudent : 3 incorrect password attempts ; TTY=tty1 ; PWD=/home/socstudent ; USER=root ; COMMAND=/usr/bin/true
f id	1787249393.71036

Figure 5. Expanded Wazuh sudo alert and decoded fields.

input.type	log
location	10.10.10.4
manager.name	wazuh-server
predecoder.hostname	linux-webserver01
predecoder.program_name	sudo
predecoder.timestamp	Aug 20 18:09:53
rule.description	Three failed attempts to run sudo
rule.firedtimes	1
rule.gdpr	IV_35.7.d, IV_32.2
rule.gpg13	7.8
rule.groups	syslog, sudo
rule.hipaa	164.312.b
rule.id	5404
rule.level	10
rule.mail	false
rule.mitre.id	T1548.003
rule.mitre.tactic	Privilege Escalation, Defense Evasion
rule.mitre.technique	Sudo and Sudo Caching
rule.nist_800_53	AU.14, AC.7
rule.pci_dss	10.2.4, 10.2.5
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	Aug 20, 2026 @ 14:09:53.396

Figure 6. Rule, MITRE ATT&CK, and compliance metadata.

Alert field	Observed value
Rule	5404 — Three failed attempts to run sudo
Level / fired times	10 / 1
Decoder	sudo
Source / destination user	socstudent / root
Command	/usr/bin/true
Location	10.10.10.4
Groups	syslog, sudo
MITRE	T1548.003 — Sudo and Sudo Caching

9. MITRE ATT&CK and Compliance Context

Mapping	Analyst interpretation
T1548.003 — Sudo and Sudo Caching	Directly relevant: sudo was used in an attempt to obtain root execution context.
Privilege Escalation	Relevant tactic because the source user requested execution as root.
Defense Evasion	Rule context only; this lab did not demonstrate evasion behavior.
NIST AU.14 / AC.7	Supports individual activity attribution and unsuccessful-login controls.
PCI DSS 10.2.4 / 10.2.5	Supports logging of invalid access attempts and authentication activity.

ANALYST JUDGMENT The alert proves a failed sudo attempt, not successful privilege escalation. Rule metadata supplies context but does not replace impact validation.

10. Triage and Final Disposition

Question	Assessment
What happened?	socstudent supplied three incorrect passwords while requesting sudo execution as root.
Was escalation successful?	No. sudo denied authentication and /usr/bin/true did not execute.
Was the activity attributable?	Yes. The log records source user, target user, TTY, directory, and command.
What was the scope?	One authorized account, one harmless command, and three failures on one endpoint.
Was persistence created?	No account, configuration, service, file, or scheduled task was changed.
Classification	True positive detection; expected benign activity in an authorized lab.

FINAL DISPOSITION TRUE POSITIVE — EXPECTED LAB ACTIVITY. Wazuh correctly detected repeated failed sudo authentication; privilege escalation did not succeed and no system impact occurred.

11. Recommended Response in a Real Environment

- Validate the user, target identity, command, host, TTY, working directory, and event time.
- Review preceding and subsequent sudo, authentication, shell, process, and file events.
- Determine whether failures are isolated, repeated, distributed, or followed by success.
- Confirm whether the user is authorized for sudo and whether the command matches job responsibilities.
- Temporarily restrict or investigate the account when attempts appear malicious or anomalous.
- Escalate immediately if failures are followed by successful root access, persistence, or defense impairment.

12. Troubleshooting and Engineering Judgment

Issue	Resolution / lesson
Initial search returned two hits	Distinguished the true /usr/bin/true failure from the later sudo grep verification record.
Exact path search returned no hits	Used broader token matching and inspected decoded data.command.
Wazuh console displayed a kernel trace	Verified the TCP/514 session and dashboard operation before considering a restart.
Command typo during evidence check	Corrected tail spelling; no system change occurred.

13. Assistance and Source Transparency

This project was completed by Ryan Bryant in his personal home lab with assistance from OpenAI ChatGPT (“Sarah”). ChatGPT helped plan the safe test, interpret results, troubleshoot event searching, locate official documentation, and edit this portfolio narrative. Ryan personally operated the VMs, entered commands, generated the authentication failures, verified the local record, investigated Wazuh, captured screenshots, and confirmed the findings. ChatGPT did not independently access or operate the lab computer.

14. Skills Demonstrated

- Linux identity and sudo analysis
- Authentication-log investigation
- Safe adversary simulation
- Wazuh rule and field interpretation
- MITRE ATT&CK contextual mapping
- Evidence-versus-impact distinction
- Incident triage and disposition
- Telemetry health validation
- Troubleshooting and query refinement

15. Employer-Facing Project Summary

RESUME BULLET Generated and investigated a controlled failed privilege-escalation event in a Wazuh SIEM lab, validating sudo telemetry, rule 5404 at Level 10, decoded user and command fields, and MITRE ATT&CK T1548.003 context while confirming no privileged execution occurred.

Interview talking points

- How I selected a harmless command and ensured authentication failed before execution.
- How I distinguished an alert from a later verification command with similar text.
- Why a failed sudo alert is not proof of successful privilege escalation.
- What follow-on telemetry would establish impact in a production investigation.

16. References and Validation Sources

[Wazuh sudo rules](#)

[Wazuh log data analysis](#)

[MITRE ATT&CK T1548.003](#)

[Ubuntu sudo manual](#)

[Ubuntu sudoers manual](#)

[OpenAI key concepts](#)